

Impact of Parameter Disentanglement on Collaborative Alignment

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Figure 1: We aim at demonstrating that parameter *sharing* and *disentanglement* could impact collaborative alignment when using AR displays. Apparent 2.5D alignment can be perceived from egocentric viewpoints (left). Additional views of other users could provide complementary information, when parameters are shared and well disentangled, to improve the 3D accuracy of alignment (right).

ABSTRACT

The interactive alignment of real and virtual content in AR is often non-trivial. Positional errors along the users view direction frequently lead to the misjudgment of the objects depth. This work takes advantage of alternative users' viewpoints in collaborative settings to mitigate these errors. Furthermore, we systematically restrict the parameters used to control the virtual contents pose and investigate the impact of sharing and disentangling such parameters. Results from this work show that alignment schemes that disentangle the control parameters improve overall alignment accuracy with a similar workload for the users and no significant increase in execution time.

Index Terms: Human computer interaction (HCI)—Mixed / augmented reality; Human computer interaction (HCI)—User studies; Human computer interaction (HCI)—Collaborative interaction.

1 INTRODUCTION

Augmented Reality (AR) has shown the potential to assist users during the performance of tasks that require the manipulation and alignment of real and virtual content. Typical scenarios present virtual content at pre-computed poses while real targets must be navigated to the equivalent pose or vice versa. However, challenges associated with perceptual issues are yet unresolved. These issues affect how an observer perceives the virtual contents depth, hindering the accurate positioning and alignment of the objects of interest.

Although automatic registration of real-to-virtual objects has proven possible [4], several factors including hardware restrictions, ergonomic considerations, or scene complexity could make this option unfeasible. In these cases, accurate manual alignment be-

comes the key to enabling the registration between the virtual and real worlds. The alignment complexity in such scenarios is expected to change based on the distance between the objects and the user. The *just noticeable difference in depth*, directly proportional to the distance from the user, suggests more difficult alignment for distant objects [2]. Several approaches including additional views [3] and virtual extensions [1] have been proposed for aiding users in mitigating these issues during manual alignment tasks in AR.

In this work, we investigate the impact of parameter sharing and disentanglement for the manipulation of virtual content in collaborative AR scenarios. We suggest parameter disentanglement schemes to access sufficient degrees-of-freedom (DoF) and reduce the total uncertainty in collaborative scenarios. The effects of disentangling the alignment are evaluated by a multi-user study, where fifteen pairs of users performed alignment tasks using four different schemes. These schemes included one baseline method and three collaborative approaches with various levels of parameter sharing.

2 MULTI-VIEW ALIGNMENT SCHEMES

We proposed four different schemes to provide users with different levels of control during the alignment tasks. These schemes, depicted in Table 1 present the following properties:

Baseline. This scheme requires a primary user to align a virtual replica of an object of interest using only their viewpoint, and without receiving any type of verbal instructions from a secondary user.

Disentangled. The disentangled scheme involves both users and considers a scenario in which the 6 DoF that control the pose of the virtual object are distributed between them. The primary user controls the 4 DoF that present the lowest uncertainties for its view (i.e., the translation in the X and Y axis and their respective orientations). The secondary user is then in charge of aligning the virtual replica in the primary user's depth axis and its corresponding orientation.

Shared. In this scenario, the restrictions imposed on the primary user are lifted, allowing them to fully control the 6 DoF of the virtual replica. In addition, the secondary user is capable to control the 4 DoF with the lowest uncertainty for its respective viewpoint.

Full. This final scheme provides both users with full control of the 6 DoF of the virtual object.

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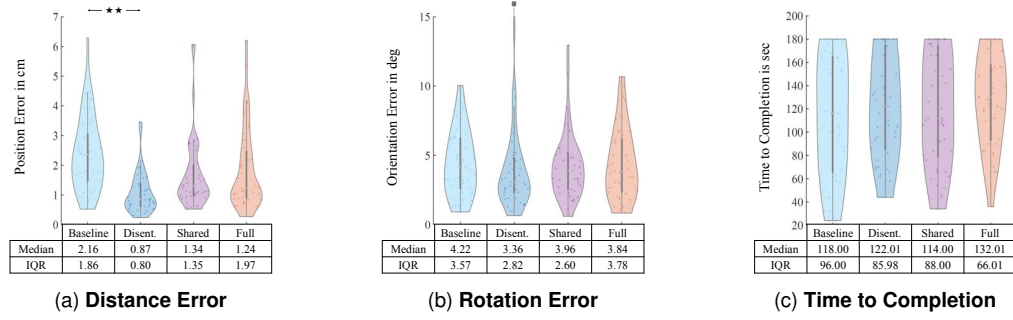


Figure 2: (a) Translation errors computed as the l-2 norm of error across all 3 axes. (b) Angular error computed as $\cos^{-1}(\frac{\text{trace}(R_e)-1}{2})$, where R_e is the associated error rotation matrix. The gray cross represents a data point of 42.7° that is not shown to enhance the visibility of the data distribution. (c) The accumulated completion time considers the time elapsed since the first interactions from the users with the virtual objects.

Scheme	Primary User			Secondary User		
	Views	DoF	Orientation	Views	DoF	Orientation
Baseline	1	x,y,z	x,y,z	-	-	-
Disentangled	1	x,y	y,z	1	z	x
Shared	1	x,y,z	x,y,z	1	y,z	x,y
Full	1	x,y,z	x,y,z	1	x,y,z	x,y,z

Table 1: Alignment schemes evaluated during our user study. The DoF are indicated using the coordinate system of the *Primary User*.

Scheme	MD	PD	TD	OP	EF	FL
Baseline	3.5±4.0	2.0±4.0	3.0±3.0	3.0±3.0	4.0±4.0	3.0±2.0
Disentangled	3.0±2.0	2.0±2.0	3.0±2.0	3.0±1.0	4.0±2.0	3.0±3.0
Shared	4.0±2.0	2.5±2.0	3.0±3.0	3.0±3.0	4.0±2.0	3.0±2.0
Full	4.0±2.0	2.0±3.0	3.5±3.0	4.0±3.0	4.0±2.0	3.0±3.0

Table 2: Median and IQR Raw-TLX scores reported by participants. MD: *Mental Demand*, PD: *Physical Demand*, TD: *Temporal Demand*, OP: *Overall Performance*, EF: *Effort*, FL: *Frustration Level*.

3 USER STUDY

Thirty participants (14 female and 16 male), aged between 21 and 29 with a mean age of 25.0 ± 2.2 years took part in the study. Participation in the study was voluntary and symptoms of, or exposure to COVID-19 were a hard exclusion criterion. All data collected from the study was anonymized. To collect reliable data, we designed a physical setup in which a non-familiar textured object from which their properties were known was rigidly attached to a 3D printed base with an optical marker (see Fig. 1).

Experimental Variables. One independent variable with four levels corresponding to the alignment schemes was considered in the study. Data regarding distance and orientation accuracy after alignment, as well as time to completion were selected as dependent variables. In addition, we collected qualitative scores using the raw values of the NASA task load index questionnaire (Raw-TLX).

4 RESULTS

Considering the non-normal distribution of the data and the within-subject nature of the study, we used Friedman tests with $\alpha = 0.05$ for the assessment of the experimental variables. Posterior Dunn-Sidak tests were used to reveal statistical significance, represented as $\star$ ($p < 0.05$), $\star\star$ ($p < 0.01$), and $\star\star\star$ ($p < 0.001$) in Fig. 2.

Regarding accuracy, Friedman tests revealed statistical significance between alignment schemes for position ($\chi^2(3) = 31.69$, $p < 0.001$). Participants achieved lower translation error using the *Disentangled* scheme ($Mdn = 0.87cm$, $IQR = 0.80cm$) when compared to the *Baseline* ($Mdn = 2.16cm$, $IQR = 1.86cm$, $p < 0.01$). Results for orientation error revealed a similar trend with the *Disentangled* scheme showing the lowest errors followed by the *Full*, *Shared*, and *Baseline*. However, no statistical significance was found between them ($\chi^2(3) = 1.20$, $p = 0.754$). Results for time to completion did not revealed statistical significance ($\chi^2(3) = 3.44$, $p = 0.328$), registering the *Shared* and *Baseline* the lowest values, followed by the *Disentangled* and *Full* schemes. Our statistical tests did not reveal statistical significance between schemes for the subjective measures. Resulting median and IQR for these metrics can be found in Table 2.

5 DISCUSSION

Overall, participants reported lower errors using the proposed collaborative schemes compared to the *Baseline*. These lower scores could be an indication that the secondary view provides more useful information than the manipulation scheme. Hypothesising that collaborative schemes require users having to come to an agreement during task performance, could explain why the *Disentangled* and *Full* schemes showed higher time to completion than the *Baseline*. One explanation to why the *Shared* presented the lowest time to completion could be the flexibility provided by the scheme, which enables skilled users to quickly align 4 of the DoF of the object while non-skilled users could focus their attention on the manipulation of the remaining 2 DoF. However, further studies with a larger population would need to be conducted in this regard.

6 CONCLUSION

This work proposed a methodology to share and disentangle DoF access to reduce total alignment uncertainty in collaborative settings. Our observations indicate that collaborative schemes can improve the overall alignment accuracy without significantly influencing the time required to complete the task. The subjective measures revealed that users can adapt to the proposed schemes without significantly increasing the workload. Finally, we hope that our theoretical findings can improve alignment and enhance the rapid adoption of AR.

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